Staggered kite finite elements for the shallow-water equations

Notes on the implementation in `dmplex.jl`

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Working notes, October 18, 2026

Abstract

These notes document the staggered kite discretization of the rotating shallow-water equations (EGU 2026 poster) as implemented in the Julia package `dmplex.jl`: the kite grid and its cell centers, the finite-element spaces and their degrees of freedom, the discrete operators, the conservation properties of the linear and nonlinear schemes, and numerical results. Open questions — the lowest-order pairing, other velocity elements, the pair of Wess et al. with Coriolis, damping and parallelization — are collected at the end.

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1 Equations

On a closed surface Ω (the unit sphere or a doubly periodic plane) with outward unit normal $\hat{\mathbf{r}}$, the rotating shallow-water equations in vector-invariant form are

$$\partial_t \mathbf{u} + q\,\hat{\mathbf{r}} \times \mathbf{F} + \nabla\big(K + gh\big) = 0, \qquad \partial_t h + \nabla \cdot \mathbf{F} = 0, \tag{1}$$

with mass flux $\mathbf{F} = h\mathbf{u}$, kinetic energy $K = \frac{1}{2}|\mathbf{u}|^2$, absolute vorticity $\omega = \zeta + f$, $\zeta = \hat{\mathbf{r}} \cdot \nabla \times \mathbf{u}$, and potential vorticity $q = \omega/h$ (so $q\,\hat{\mathbf{r}} \times \mathbf{F} = \omega\,\hat{\mathbf{r}} \times \mathbf{u}$). The equations conserve mass $\int h$ and energy $E = \int \frac{1}{2}h|\mathbf{u}|^2 + \frac{1}{2}gh^2$. For the sphere we use units with radius $a = 1$ and rotation rate $\Omega = 1$, so $f = 2z$.

2 The kite grid

Construction. Let $\mathcal{P}$ be a polygonal primal mesh (triangles, quadrilaterals, the hexagons and pentagons of a Voronoi mesh). Every cell $P \in \mathcal{P}$ with vertices $w_1, \ldots, w_n$ (counter-clockwise seen from outside) is split into n quadrilaterals (*kites*)

$$K_i = (w_i,\ m_i,\ c_P,\ m_{i-1}), \qquad i = 1, \ldots, n,$$

where c_P is the cell center and m_i a point on the primal edge (w_i, w_{i+1}). The kites sharing a primal vertex w form the dual cell $\widetilde{P}_w$. In `create_kite_mesh` every kite has the canonical local numbering: vertex 1 is the primal vertex, vertex 3 the center; local edges 1 and 4 are halves of primal edges (*half edges*), local edges 2 and 3 join an edge point to the center (*interior edges*); the interior edges of P are exactly the boundaries of the dual cells. Labels record the parent cell, the dual cell and the vertex and edge types.

Cell centers. The center is chosen so that it coincides with the vertex of the orthogonal dual mesh whenever there is one (`cell_centers`):

- triangles: the circumcenter (vertex of the Voronoi dual);
- Voronoi meshes: the generators. Every Voronoi edge with great-circle normal $\mathbf{n}_e = a \times b$ is the perpendicular bisector of the generators g_i, g_j of its two cells,

  $$g_i - g_j \parallel \mathbf{n}_e, \qquad \mathbf{n}_e \cdot (g_i + g_j) = 0, \tag{2}$$

  which is a homogeneous linear system for all generators. Its least-squares null vector is computed by sparse inverse iteration. For MPAS meshes this recovers `xCell` to 10^{-14}, for the ICON dual `vlon/vlat` to 10^{-14};
- other meshes (cubed sphere, HEALPix, planes): the centroid.

On meshes with an orthogonal dual, m_i is the crossing of the primal edge with the dual edge $(c_P, c_{P'})$, so the interior kite edges lie on the dual edges and each dual cell is exactly the dual polygon (the Delaunay triangle of an MPAS mesh); otherwise m_i is the edge midpoint. The number of kites equals $\sum_P n_P$, e.g. 245 760 for the MPAS mesh `x1.40962`.

3 Finite-element spaces

All kite spaces live on the reference square $[-1, 1]^2$ with the primal vertex at $(-1, -1)$ and the center at $(1, 1)$. The reference map is the bilinear interpolant of the four kite vertices, radially projected onto the sphere (curved elements); a chordal variant without projection is available.

Height: `KiteRadau(p)`. Q_p on every kite with nodes at the tensor Gauss–Radau points $r_0 < \cdots < r_p = 1$ that include the end point at the center side. Nodes lie in the kite interior (p^2), on the interior edges (p each) and at the center (one), never on primal edges. Hence h is continuous inside a primal cell and broken across primal edges,

$$V_h = \{h \in L^2(\Omega) : h|_P \in H^1(P)\ \forall P \in \mathcal{P},\ h|_K \in Q_p\},$$

with $1+n_P(p+p^2)$ degrees of freedom per primal cell. With the Radau nodes as quadrature the mass matrix is diagonal; the rule is exact for degree $2p$ per direction, so for $p \geq 1$ the lumped mass integrates the bilinear area element exactly and is consistent. For $p = 0$ the single Radau node sits at the center corner and 4 $\det J(c)$ is an $O(1)$ wrong kite area (kites do not become parallelograms under refinement); since h is then constant per primal cell the exact mass matrix (the cell area) is used instead.

Velocity: `KiteBrokenRT(k)`. RTCF_k on every kite with normal continuity across half edges (inside a dual cell) and none across interior edges (the dual cell boundaries):

$$V_u = \{\mathbf{u} \in L^2(\Omega)^2 : \mathbf{u}|_{\widetilde{P}} \in H(\mathrm{div}, \widetilde{P})\ \forall \widetilde{P} \in \widetilde{\mathcal{P}},\ \mathbf{u}|_K \in \mathrm{RTCF}_k\}.$$

Edge degrees of freedom are normal components at the $k+1$ Gauss points; the degrees of freedom of the two interior edges are stored with the kite itself. The exact mass matrix is block diagonal with one block per dual cell (9 for RT0 on a triangular dual cell, 12 for a quadrilateral one, 30 for RT1 on a triangular dual cell).

Other velocity spaces. Two alternatives with the same continuity pattern (normal components continuous across half edges, broken across interior edges) are implemented:

- `KiteBrokenBDM(k)`: BDM_k on every kite, $[P_k]^2 \oplus \mathrm{span}\{\mathrm{curl}(x^{k+1}y), \mathrm{curl}(xy^{k+1})\}$, normal components at the $k+1$ Gauss points of each edge and, for $k \geq 2$, moments against $[P_{k-2}]^2$; 8 instead of 12 degrees of freedom per kite for $k = 1$. The interior-edge degrees of freedom are stored with the kite as for RT.
- `KiteDualRadau(p)`, the velocity space of the dual cell method of Wess, Kapidani, Codecasa and Schöberl (2024): $Q_p \times Q_p$ on every kite with nodes at the tensor Gauss–Radau points that include the primal-vertex side (the mirror image of the height nodes). On the half edges the degrees of freedom are the normal components, shared by the two kites of the dual cell; inside the kite the component values. There are no nodes on the interior edges, so the field is broken there without any special storage. With the nodes as quadrature the mass matrix couples only the components at one node: blocks of size 3 for $p = 1$ on hexagon kites, independent of the mesh.

Vorticity. Continuous Q_{k+1} (`Lagrange(k+1)`) on the kites with Gauss–Lobatto nodes and lumped, diagonal mass.

Implementation. A `Section` assigns degrees of freedom to mesh points (vertices, edges, cells). Layouts that depend on point labels extend `num_nodes(layout, dm, p)`, and an element may store dofs on another point than the one they belong to geometrically (the interior-edge RT dofs on the kite). The closure of a kite then delivers the global indices, with node order reversed and signs flipped on reversed edges, so the spaces are conforming where intended; the tests check the continuity pattern across both edge types to machine precision.

4 Discrete operators

With the contravariant Piola map $\mathbf{v} = J\hat{\mathbf{v}}/\det J$ one has $\mathbf{v} \cdot \nabla\eta\, dA = \hat{\mathbf{v}} \cdot \hat{\nabla}\hat{\eta}\, d\xi$, $(\nabla \cdot \mathbf{v})\, \eta\, dA = (\hat{\nabla} \cdot \hat{\mathbf{v}})\, \hat{\eta}\, d\xi$ and $\mathbf{v} \cdot \mathbf{n}\, ds = \hat{\mathbf{v}} \cdot \hat{\nu}\, dt$ with the scaled reference edge normal $\hat{\nu}$; both operators below are therefore independent of the metric.

Divergence. For $\eta \in V_h$, $\mathbf{F} \in V_u$,

$$D(\eta; \mathbf{F}) = \sum_K \int_K \mathbf{F} \cdot \nabla \eta - \sum_K \sum_{e \subset \partial K \cap \partial \mathcal{P}} \int_e \eta_K \, \mathbf{F} \cdot \mathbf{n}_K, \tag{3}$$

summed over the half edges of each kite with the trace of η from inside K; as a matrix $B_{ij} = D(\eta_i; \mathbf{v}_j)$.

Gradient. For $p \in V_h$, $\mathbf{v} \in V_u$,

$$G(p; \mathbf{v}) = \sum_K \int_K p \, \nabla \cdot \mathbf{v} - \sum_K \sum_{e \subset \partial K,\ e \text{ interior}} \int_e p \, \mathbf{v}_K \cdot \mathbf{n}_K, \tag{4}$$

where p is single valued on interior edges.

Proposition 1. *$G(p; \mathbf{v}) = -D(p; \mathbf{v})$, i.e. $G = -B^T$, and $\mathbf{1}^T B = 0$.*

Proof. Integrating by parts on each kite, $\int_K p \, \nabla \cdot \mathbf{v} = -\int_K \nabla p \cdot \mathbf{v} + \int_{\partial K} p \, \mathbf{v} \cdot \mathbf{n}_K$. The boundary integral splits into half edges and interior edges; the interior edges cancel against the face term of G, leaving $-\sum_K \int_K \nabla p \cdot \mathbf{v} + \sum_K \sum_{\text{half}} \int p_K \, \mathbf{v} \cdot \mathbf{n}_K = -D(p; \mathbf{v})$. For $\eta \equiv 1 \in V_h$, $D(1; \mathbf{v}) = -\sum_K \sum_{\text{half}} \int \mathbf{v} \cdot \mathbf{n}_K = 0$, since $\mathbf{v} \cdot \mathbf{n}$ is continuous across half edges and both sides appear with opposite normals. □

Both matrices are assembled independently in the code; the test suite checks $\|G + B^T\|_\infty < 10^{-12}$.

Vorticity, kinetic energy, Coriolis. $\langle \zeta, \gamma \rangle = -\int \mathbf{u} \cdot (\hat{\mathbf{r}} \times \nabla \gamma)$ for $\gamma \in Q_{k+1}$ (lumped mass), $\langle K, \eta \rangle = \int \frac{1}{2} |\mathbf{u}|^2 \eta$ (lumped mass of V_h), and $C_{ij} = \int f \, \mathbf{v}_i \cdot (\hat{\mathbf{r}} \times \mathbf{v}_j)$, which is antisymmetric. The surface normal $\hat{\mathbf{r}}$ is computed from the Jacobian, $\hat{\mathbf{r}} = J_1 \times J_2 / \det J$, which is $x/|x|$ on curved elements and the element normal on chordal ones.

5 Semi-discrete schemes and conservation

Linear.

$$M_h \dot{h} = H \, B \, \mathbf{u}, \qquad M_u \dot{\mathbf{u}} = -C\mathbf{u} - g \, B^T h. \tag{5}$$

Mass $\mathbf{1}^T M_h h$ is conserved because $\mathbf{1}^T B = 0$, energy $\frac{1}{2} H \mathbf{u}^T M_u \mathbf{u} + \frac{1}{2} g h^T M_h h$ because C is antisymmetric and the gradient is the negative adjoint of the divergence.

Nonlinear, poster form.

$$\langle \dot{h}, \eta \rangle = \sum_K \int_K h \mathbf{u} \cdot \nabla \eta - \sum_K \sum_{\text{half}} \int \eta_K \, h_K \, \mathbf{u} \cdot \mathbf{n}_K, \tag{6}$$

$$\int \dot{\mathbf{u}} \cdot \mathbf{v} = -\int (\zeta + f) \, (\hat{\mathbf{r}} \times \mathbf{u}) \cdot \mathbf{v} + G(gh + K; \mathbf{v}), \tag{7}$$

with h_K the trace from inside the kite. On primal edges h is two valued, so the flux $h_K \, \mathbf{u} \cdot \mathbf{n}$ is not single valued: neither mass nor energy is conserved exactly.

Nonlinear, conservative variant (default). Project the mass flux, $\mathbf{F} = \Pi_{V_u}(h\mathbf{u})$, i.e. $M_u\mathbf{F} = \left(\int h\mathbf{u}\cdot\mathbf{v}_i\right)_i$ (one solve with the block-diagonal velocity mass), and use the potential vorticity $q = (\zeta + f)/h$ at the quadrature points:

$$M_h\dot{h} = B\mathbf{F}, \qquad \int \dot{\mathbf{u}}\cdot\mathbf{v} = -\int q\,(\hat{\mathbf{r}}\times\mathbf{F})\cdot\mathbf{v} - B^T(gh+K). \tag{8}$$

Proposition 2. *The conservative variant conserves mass and the discrete energy* $E_h = \int \frac{1}{2}h|\mathbf{u}|^2 + \frac{1}{2}g\,h^T M_h h$ *exactly.*

Proof. Mass: $\mathbf{1}^T M_h\dot{h} = \mathbf{1}^T B\mathbf{F} = 0$. Energy: with the same quadrature in E_h and in the right-hand sides, $\partial E_h/\partial\mathbf{u} = M_u\mathbf{F}$ and $\partial E_h/\partial h = M_h(K+gh)$. Hence $\dot{E}_h = \mathbf{F}^T M_u\dot{\mathbf{u}} + (K+gh)^T M_h\dot{h} = -\int q\,(\hat{\mathbf{r}}\times\mathbf{F})\cdot\mathbf{F} - \mathbf{F}^T B^T(gh+K) + (gh+K)^T B\mathbf{F} = 0$, since $(\hat{\mathbf{r}}\times\mathbf{F})\cdot\mathbf{F} = 0$ pointwise. □

This is the framework of McRae and Cotter (2014) transferred to the kite spaces; all mass matrices stay diagonal or block diagonal. The test suite checks $\dot{E}_h$ and $\mathbf{1}^T M_h\dot{h}$ to round-off on hexagonal, triangular and quadrilateral kites and on periodic planes.

6 Numerical results

Williamson test case 2 (steady geostrophic flow, $u_0 = 0.2$, $g = 1$, $T = 1$) measures $\|h(T) - h(0)\|_{M_h}$; kites of the hexagonal (Voronoi) meshes `hex(r)` have 960, 3840 and 15360 kites for $r = 2, 3, 4$.

6.1 Lowest order

L^2 errors of the discrete gradient $M_u^{-1}B^T\Pi p$ and divergence $M_h^{-1}B\,\Pi\mathbf{u}$ against the projections of the exact operators for smooth p and $\mathbf{u}$ (hexagon kites with 960, 3840, 15360 kites; cubed-sphere kites with 384, 1536, 6144 kites):

pairing	mesh	gradient error	divergence error
$Q_0 \times \mathrm{RT}_0$	hex	7.4, 7.6, 7.7	0.017, 0.0069, 0.0030
	cube	5.6, 5.8, 5.9	0.065, 0.076, 0.067
$Q_1 \times \mathrm{RT}_0$	hex	0.10, 0.052, 0.026	0.23, 0.21, 0.21
	cube	0.19, 0.094, 0.047	0.40, 0.36, 0.31
$Q_1 \times \mathrm{RT}_1$	hex	0.21, 0.11, 0.052	0.016, 0.0060, 0.0027
	cube	0.31, 0.15, 0.077	0.044, 0.022, 0.012
$Q_2 \times \mathrm{RT}_1$	hex	0.0038, 0.00091, 0.00023	0.088, 0.045, 0.023
	cube	0.0091, 0.0023, 0.00057	0.059, 0.035, 0.019

At lowest order one of the two operators is inconsistent. With Q_0 heights the pressure is constant per primal cell and only sets the half-edge normal components of the gradient; the kite-local interior-edge velocity degrees of freedom are not controlled by the pressure. With $Q_1 \times \mathrm{RT}_0$ the discrete divergence of the broken RT0 field tested with Q_1 does not converge. From RT_1 on both operators converge. This agrees with the experience with CGDycore that the lowest order does not work.

6.2 Linear model

$Q_1 \times \mathrm{RT}_1$, RK4, geostrophic steady state:

primal mesh	coarse	medium	fine
hexagons (162, 642, 2562 cells)	$3.09 \cdot 10^{-3}$	$7.72 \cdot 10^{-4}$	$1.94 \cdot 10^{-4}$
triangles (80, 320, 1280)	$9.67 \cdot 10^{-3}$	$2.46 \cdot 10^{-3}$	$6.16 \cdot 10^{-4}$
cubed sphere (96, 384, 1536)	$7.88 \cdot 10^{-3}$	$1.94 \cdot 10^{-3}$	$4.88 \cdot 10^{-4}$

Second order on all meshes; mass to 10^{-15}, energy up to the RK4 error.

6.3 Nonlinear model

Williamson test case 2, RT1 velocity, CG2 vorticity, SSP-RK3:

variant	h	hex(2)	hex(3)	mass / energy
poster form	Q_1	$2.05 \cdot 10^{-3}$	$8.76 \cdot 10^{-4}$	$\dot{M} \approx 10^{-4}$, $\Delta E/E \approx 10^{-4}$
poster form	Q_2	$5.9 \cdot 10^{-4}$	$5.2 \cdot 10^{-3}$ (grows)	not conserved
projected flux	Q_2	$3.60 \cdot 10^{-4}$	$8.79 \cdot 10^{-5}$	exact / 10^{-12}
projected flux + q	Q_2	$3.60 \cdot 10^{-4}$	$8.80 \cdot 10^{-5}$	exact / exact

The pair RT1 / Q_2 / CG2 is the most accurate and, with the projected flux, second-order convergent. With the poster's own-trace flux, Q_2 heights grew on the finer mesh; nothing damps the jumps of h across primal edges there.

Quadrature. All nonlinear terms use one Gauss rule per kite with precomputed metric terms ($J/\det J$, dA, $\hat{\mathbf{r}}$, f). Errors on hex(2) / hex(3) and cost per SSP-RK3 step on hex(3):

rule	hex(2)	hex(3)	ms/step
3×3	$3.606 \cdot 10^{-4}$	$8.765 \cdot 10^{-5}$	42
4×4	$3.601 \cdot 10^{-4}$	$8.796 \cdot 10^{-5}$	53
5×5	$3.601 \cdot 10^{-4}$	$8.796 \cdot 10^{-5}$	68

The minimal rule (3×3 for RT1) is the default; energy conservation is an algebraic property and holds for every rule. A time step does not allocate.

Curved versus chordal elements. With precomputed metric terms both cost the same; chordal (flat bilinear) kites give $9.10 \cdot 10^{-4}$ and $2.50 \cdot 10^{-4}$ (rate 1.87) instead of $3.60 \cdot 10^{-4}$ and $8.80 \cdot 10^{-5}$ (rate 2.03). Curved elements are the default.

6.4 Velocity spaces: RT, BDM and the pair of Wess et al.

Operator consistency (L^2 errors of the discrete gradient and divergence on hexagon kites with 960, 3840, 15360 kites) and the linear geostrophic steady state with Coriolis ($\|h(1) - h(0)\|$, RK4, finest mesh and rate):

pair (velocity / height)	gradient	divergence	geostrophic, finest (rate)
RT1 / Q_1	0.21, 0.11, 0.052	0.016, 0.0060, 0.0027	$1.94 \cdot 10^{-4}$ (1.99)
RT1 / Q_2	0.0038, 0.00091, 0.00023	0.088, 0.045, 0.023	$4.12 \cdot 10^{-5}$ (2.10)
BDM1 / Q_1	0.13, 0.065, 0.032	0.28, 0.26, 0.25	$1.46 \cdot 10^{-4}$ (1.78)
BDM1 / Q_2	0.0021, 0.00048, 0.00012	0.84, 0.83, 0.82	$1.61 \cdot 10^{-4}$ (1.20)
Wess $p = 1$ (Q_1^2 / Q_1)	0.14, 0.069, 0.035	0.21, 0.18, 0.18	$1.68 \cdot 10^{-4}$ (1.86)
Wess $p = 2$ (Q_2^2 / Q_2)	0.0070, 0.0018, 0.00044	0.026, 0.013, 0.0064	$5.61 \cdot 10^{-6}$ (2.07)

The divergence error of BDM1 and of the Wess pair with $p = 1$ does not converge in this norm (the discrete divergence of the broken velocity tested with the height space), yet the geostrophic steady state converges for all pairs. The Wess pair with $p = 2$ is the most accurate, about seven times better than RT1/Q_2; energy is conserved exactly by all pairs.

Wess pair with Coriolis. Assembling the Coriolis term with the velocity nodes as quadrature (lumped) instead of exactly makes no difference ($1.68 \cdot 10^{-4}$ versus $1.63 \cdot 10^{-4}$ for $p = 1$, $5.61 \cdot 10^{-6}$ versus $5.62 \cdot 10^{-6}$ for $p = 2$ on the finest mesh, same rates). In the nonlinear Williamson test case 2 (conservative variant, SSP-RK3) the Wess pair with $p = 2$ gives $8.54 \cdot 10^{-5}$ and $2.02 \cdot 10^{-5}$ on hex(2), hex(3) (rate 2.08) against $3.61 \cdot 10^{-4}$, $8.77 \cdot 10^{-5}$ for RT1/Q_2, with mass and energy conserved; with $p = 1$ it converges with rate 1.4. In this implementation the pair therefore works with Coriolis and in the nonlinear case. Differences to a failing implementation could lie in the face terms (here the kite divergence B and $G = -B^T$, both checked independently), in the orientation of the shared normal components at the half edges, or in the nonlinear terms (here the conservative variant with projected flux).

6.5 Kite method versus the standard mixed setup on the kite quads

The kite grid is an ordinary quadrilateral mesh, so the compatible pair RT1 × DG1 × CG2 of McRae and Cotter (global, non-diagonal mass matrices, `NonlinearShallowWater`) runs on it unchanged. Williamson test case 2, SSP-RK3, same time step:

mesh	method	u dofs	h dofs	error	mass solve	ms/step
hex(2)	kite RT1/Q_2/CG2	9600	5922	$3.61 \cdot 10^{-4}$	blocks (30)	6.9
	mixed RT1/DG1/CG2	7680	3840	$1.94 \cdot 10^{-4}$	Cholesky	20.0
hex(3)	kite RT1/Q_2/CG2	38400	23682	$8.77 \cdot 10^{-5}$	blocks (30)	38.4
	mixed RT1/DG1/CG2	30720	15360	$4.72 \cdot 10^{-5}$	Cholesky	98.5

Both are second order and conserve energy; the mixed setup is about 1.9× more accurate with fewer degrees of freedom. The timings are not a fair comparison of the methods: the mixed model in `dmplex.jl` still uses the older allocating kernels, and its global mass solves (sparse Cholesky) are the part that the kite method replaces by local block inverses, which matters most for explicit time stepping, on GPUs and in parallel.

6.6 Galewsky jet

2562 hexagons, 15360 kites ($\approx 182\,\mathrm{km}$), $\Delta t = 66.9\,\mathrm{s}$, 6 days: the jet becomes unstable after day 3 and rolls up by day 6; mass changes by $6 \cdot 10^{-12}$, energy by $1.2 \cdot 10^{-8}$ (time integration only). Grid-scale noise is visible near the vortices (no dissipation).

6.7 Bickley jet

Doubly periodic $[0, 4\pi)^2$, $u = \mathrm{sech}^2(y - 2\pi)$, no rotation, Froude number 0.2, perturbation in wavenumber 1; 32×32 quadrilaterals, 4096 kites, $t = 40$: vortex street with two cat's-eye pairs; mass $8 \cdot 10^{-11}$, energy $8 \cdot 10^{-10}$.

7 Open questions and proposals

7.1 Other velocity elements

Broken BDM and the space of Wess et al. are implemented (Section 3 and the results). Open: BDM2 and $p \geq 3$ for the Wess pair, the dual-cell method on triangles only (Wess et al. use

triangular primal meshes) versus general polygons, and a lumped version of the RT velocity mass.

7.2 Damping

For comparison, Gusto's standard shallow-water model (`SIQNModel`) obtains its dissipation implicitly from upwinded DG transport (`DGUpwind`) of the depth and of the velocity in vector-invariant form, with SSP-RK3 transport inside a semi-implicit quasi-Newton step; interior-penalty diffusion and SUPG are optional. Options for the kite method, in order of preference:

1. *APVM* (anticipated potential vorticity): replace q by $q - \tau\,\mathbf{u}\cdot\nabla q$, $\tau = \alpha\Delta t$, in the vorticity term. The term stays of the form $q^*\,\hat{\mathbf{r}}\times\mathbf{F}$, so energy is still conserved exactly while potential enstrophy is dissipated (McRae and Cotter 2014). ∇q is available from $\nabla\zeta$ (CG2) and ∇h inside the kite.

2. *Upwind height flux on primal edges*, where h jumps: replace the own-trace h_K by the upwind trace in the poster form. The flux is single valued, so mass is conserved, and energy is dissipated by the jumps of h (the analogue of Gusto's upwinded depth transport).

3. *Biharmonic hyperviscosity* on the velocity, $\Delta\mathbf{u} = \nabla\delta - \hat{\mathbf{r}}\times\nabla\zeta$ with the discrete divergence $\delta = M_h^{-1}B\mathbf{u}$ and vorticity ζ, applied twice, $\nu_4 \propto \Delta x^4/\tau_4$; as in MPAS and CGDycore, mainly for comparison.

7.3 Comparison with the standard mixed setup on the kite quads

The kite grid is an ordinary quadrilateral mesh, so the standard compatible pair $\mathrm{RT}_k\times\mathrm{DG}_k\times\mathrm{CG}_{k+1}$ (`NonlinearShallowWater` on the kite `Plex`) can be run on it directly; its mass matrices are not block diagonal. See the results section.

7.4 Parallelization

Kernels. The nonlinear terms are KernelAbstractions kernels over a list of kites: gather the degrees of freedom, evaluate at the quadrature points with the stored metric terms, and scatter with atomic adds (Atomix). On the CPU backend the kernels run on the Julia threads, on a GPU unchanged; the scatter order and therefore the round-off depend on the scheduling. The block-diagonal mass solves need no atomics since the blocks are disjoint. (The alternative to atomics is colouring: kites of one colour share no degrees of freedom, colours are processed one after the other.)

Distribution (MPI). The primal cells are partitioned with the zonal equal-area regions of Leopardi (Atlas `EqualRegionsPartitioner`, CGDycore `DecomposeEqualArea`); kites follow their primal cell. Each rank holds one vertex ring of ghost primal cells. Ownership: height dofs follow their primal cell, velocity dofs their dual-cell block (the smallest part around the dual vertex), vorticity dofs the smallest part among the kites touching them. With the ghost ring every owned dof has all its kites locally, so a rank computes the right-hand sides over its owned and ghost kites, and they are complete for its owned dofs; all mass blocks of owned dofs are local. Kites and degrees of freedom are numbered locally (owned first, then ghosts); the global numbering is only used during setup and to gather states. Per tendency evaluation the halo exchange updates the velocity $\mathbf{u}$ and height h (prognostic), the vorticity ζ and, in the conservative variant, the mass flux $\mathbf{F}$ (diagnostic). The kinetic energy is not exchanged: it lives in the height space, whose ghost degrees of freedom belong to ghost primal cells whose kites are all local. The linear operators are applied to the owned rows only. The runs on one and on three ranks agree to 10^{-12}, and mass and energy are conserved across ranks. The global mesh is still built on every rank during setup.

First scaling test. Galewsky setup, 2562 hexagons (15360 kites), Mac with 8 performance and 8 efficiency cores, warmed-up SSP-RK3 steps; per tendency the maximum over ranks:

ranks	ms/step	kernels	halo (4×)	sparse products	block solves
1	241	34.7	0.9	5.5	19.8
2	145	21.0	1.6	4.4	10.5
4	90	14.6	1.7	3.1	6.2
8	75	13.0	3.6	2.8	4.9

Up to four ranks the speed-up is reasonable (2.7); from four to eight ranks the kernels hardly get faster, although each rank has half the kites. The per-kite cost grows by a factor of 2.4, which points to the global-length vectors (poor locality of the gathers) and to ranks on efficiency cores. With the local numbering 1, 2 and 4 ranks took 205, 129 and 82 ms per step; a repeated test on a loaded machine (load average 9–11) was not meaningful, so the scaling beyond four ranks still has to be measured on an idle machine.